D1.1 DNA replication

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 STATE three reasons why cells replicate their DNA. [D1.1.1] [3]

2 A template strand has the sequence A C G G T A. WRITE the complementary strand. [D1.1.2] [1]

3 STATE the type of bond broken by helicase. [D1.1.3] [1]

4 STATE the approximate temperature of each of the three steps in a PCR cycle. [D1.1.4] [2]

5 STATE the electrode that DNA moves towards in gel electrophoresis. [D1.1.4] [1]

Section B: Understanding

12 MARKS

6 EXPLAIN what is meant by semi-conservative replication. [D1.1.2] [2]

7 EXPLAIN how complementary base pairing allows DNA to be copied accurately. [D1.1.2] [3]

8 OUTLINE what helicase and DNA polymerase do in replication. [D1.1.3] [3]

9 **EXPLAIN** why Taq polymerase is used in PCR, and not a polymerase from a human cell. [D1.1.4] [2]

10 **EXPLAIN** why short DNA fragments travel further through a gel than long fragments. [D1.1.4] [2]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D1.1.4]

A PCR started with 200 copies of a DNA section. Each cycle took 4 minutes.

NUMBER OF CYCLES	0	1	2	3	10
COPIES OF THE SECTION	200	400	800		

a **CALCULATE** the number of copies after 3 cycles and after 10 cycles. [2]

b **CALCULATE** the time taken for 10 cycles. [1]

c **SUGGEST** one reason why real reactions stop doubling after many cycles. [1]

12 [D1.1.5]

The table shows the bands found on a gel, at four loci, for a mother, her child and two men. A tick shows that the band is present.

PERSON	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
MOTHER	✓		✓		✓	
CHILD	✓	✓		✓	✓	
MAN R		✓		✓		✓
MAN S	✓		✓	✓		

a **IDENTIFY** the bands that the child must have inherited from the father. [2]

b **DEDUCE** which man could be the father, giving a reason. [2]

c **EXPLAIN** why many loci are compared and not just one. [2]

d **EXPLAIN** why PCR must be carried out before electrophoresis when a sample is very small. [2]

e **STATE** what is run in one lane to estimate the length of the fragments. [2]

Section D: Longer answers

6 MARKS

13 **DESCRIBE** how a small sample of DNA found at a crime scene could be used to identify a suspect. [6]

[D1.1.4, D1.1.5]